

Architecture and Preprocessing Choices under Dataset Shift in Single-Channel EEG Sleep Staging: A Controlled Evaluation on Sleep-EDF and SHHS1

Ngo Nhat Quan^{1,*} and Pham Thai Son¹

¹*Faculty of Computer Science, University of Information Technology,
Vietnam National University Ho Chi Minh City, Ho Chi Minh City, Viet Nam*

*First and corresponding author: 23521258@gm.uit.edu.vn
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Abstract

Background. Cross-cohort losses are difficult to attribute when architecture and preprocessing change together. We tested which pipeline choices remained useful before cross-cohort adaptation.

Methods. Six configurations were compared on Sleep-EDF Expanded (78 subjects) under shared 10-fold partitions and selection rules. E1 changed only the sequence model; E2 replaced E1's 75-dimensional C/P/N feature-context package with a 128-dimensional current-epoch ResNet-1D encoder. Paired bootstrap and Wilcoxon–Holm quantified uncertainty. Transfer used 180 locked SHHS1 participants without weight updates. E6 used statistics from each complete unlabelled target recording and is therefore label-free but transductive.

Results. E3 attained the highest Sleep-EDF macro-F1 (0.7904) and exceeded E6 by 0.0213 (CI_{95%} [0.0122, 0.0307]; Holm $p = 0.0012$); a second seed preserved the direction but not Holm significance. Other contrasts were small and non-significant. E3 provided a 3.76× forward-pass speed-up over the CNN–BiLSTM control, with 4.37× as many parameters and 28.4% more peak memory. On SHHS1, E3 exceeded the control by 0.0412 subject-mean macro-F1 (CI_{95%} [0.0314, 0.0512]) and E6 by 0.0274. E0 and E3 nevertheless misclassified 72.3% and 73.1% of N3 epochs as N2 and had similar N3 recall (0.2610 and 0.2582); E6 reduced recall to 0.2005.

Conclusions. The substitutions offered operational but no stable predictive gains. In this post-hoc comparison, preprocessing was the stronger development axis. N2–N3 and N2–REM should be prioritised for labelled adaptation; target-record normalisation alone was insufficient.

Keywords: sleep staging; single-channel EEG; temporal convolutional network; domain shift; target-domain adaptation; paired inference.

1. Introduction

Sleep staging assigns one of five labels—W, N1, N2, N3 and REM—to every 30 s epoch of a polysomnographic recording. Manual scoring is slow, requires trained expertise, and retains irreducible disagreement between scorers; in the inter-scorer reliability programme of the American Academy of Sleep Medicine, N1 shows the lowest agreement of all stages [1]. An automated system therefore has to be judged not by overall accuracy

alone, which is dominated by the most frequent stages, but by class-balanced metrics, by stability across individuals, and by its behaviour when the recording device, montage or population changes.

Progress in this field is difficult to read from published tables. Studies differ simultaneously in dataset version, subject partitioning, wake-trimming policy, filtering, normalisation, sequence length and model-selection rule, so a reported improvement cannot be attributed to the architecture that the paper nominally proposes. This creates a concrete development problem: before collecting target-domain labels, a practitioner cannot tell whether a new encoder, a new sequence model or a change in signal handling is the more defensible use of effort. A comparison in which one component changes at a time is needed to separate these decisions.

We take ZleepAnlystNet [2], a two-stage single-channel model that couples a bank of class-specific convolutional encoders to a BiLSTM, as an internal control. E1 isolates the replacement of the BiLSTM by a temporal convolutional network (TCN) [3]. E2 is not an isolated encoder contrast: it replaces E1’s 75-dimensional current/previous/next (C/P/N) feature-context package with a 128-dimensional current-epoch one-dimensional residual encoder [4]. We additionally treat signal processing as an experimental factor because filtering and record-wise normalisation can change the information available to a downstream classifier.

The study is organised around three deployment-oriented questions. (i) Under a shared subject-wise protocol, do the isolated TCN substitution and bundled E2 feature/context replacement provide benefits large and stable enough to justify changing the pipeline? (ii) Which evaluated pipeline and preprocessing choices retain their benefit under label-free cross-cohort transfer to an independent cohort with a different EEG derivation and age distribution? (iii) When transfer performance falls, which class transitions account for the loss and do they point to a generic preprocessing remedy or to targeted adaptation?

Our contributions are the following.

- (1) A leakage-controlled 10-fold subject-wise protocol with an isolated sequence-model contrast, a clearly labelled bundled feature/context replacement, and preprocessing contrasts under shared partitions and model-selection rules.
- (2) A locked cross-cohort evaluation on 180 SHHS Visit-1 participants, showing that the ranking and magnitude of design choices can differ between internal validation and cross-cohort use.
- (3) A class- and confusion-channel analysis showing that under-detection of N3 as N2 recurs in both the 15-CNN–BiLSTM control and ResNet-1D–TCN, while over-detection of REM from N2 is a second E3 priority. The accompanying E6 sensitivity analysis shows that per-record normalisation alone does not resolve the cross-model N3 failure.
- (4) A quantified operational trade-off: the implemented ResNet-1D–TCN pipeline reduces measured forward latency and training-and-validation wall-clock time relative to the 15-branch CNN–BiLSTM control, while increasing parameter count and peak memory and without establishing predictive superiority.

Throughout, we distinguish pre-specified comparisons, secondary evidence and post-hoc contrasts, and we confine claims to the evidence class that supports them.

2. Related work

DeepSleepNet combined a dual-branch CNN with a BiLSTM to capture epoch morphology and temporal dependence [5]. AttnSleep introduced attention over multi-resolution features [6], and SleepTransformer applied self-attention with explicit uncertainty quantification [7]. Together these establish that sequential context matters substantially. They also illustrate the comparability problem: differences in dataset version, subject split and preprocessing between these works are large enough that cross-table comparison cannot substitute for an ablation carried out inside one protocol, a point emphasised in recent reviews of the field [8].

Single-channel systems also span materially different input representations. SleepInceptionNet systematically compared raw, spectral and time–frequency inputs and selected a continuous-wavelet, central-channel Inception model that scores one epoch at a time [9]. That study reinforces two points relevant here: a single EEG channel can support five-stage scoring, and preprocessing choice is part of the model definition rather than a neutral implementation detail.

Cross-dataset sleep staging is already an active methodological problem. For example, He et al. [10]

trained a domain-adaptation network across Sleep-EDF, SHHS1 and the PhysioNet Challenge data. ADAST instead uses domain-specific attention, adversarial alignment and iterative pseudo-label self-training on unlabelled target data [11]. These approaches adapt model representations or decision functions using the target domain; our label-free E6 condition only estimates recording-level normalisation statistics and does not implement domain adaptation. Van Der Donckt et al. [12] further showed that increasingly complex deep architectures need not outperform a carefully designed conventional pipeline under a matched protocol. Our study addresses the decision immediately before a labelled adaptation study: which existing pipeline choices remain useful, and which class boundaries should receive limited target-domain labels?

Sleep-EDF Expanded is the standard public benchmark for single-channel EEG [13, 14]. The Sleep Heart Health Study (SHHS) is a large multi-centre cohort assembled to study sleep-disordered breathing and cardiovascular risk [15], distributed through the National Sleep Research Resource (NSRR) [16]. The two differ in age, comorbidity, recording hardware, montage and label distribution. This makes SHHS a considerably more realistic transfer test than a held-out split of the same corpus, and it is the reason we treat the transfer result—not the in-domain ranking—as the principal evidence about deployability.

Two threads of the literature bear directly on our findings. First, the sleep slow oscillation has a pronounced anterior amplitude maximum and propagates as a travelling wave [17], so the microvolt amplitude of slow-wave activity is derivation-dependent; and slow-wave amplitude declines markedly with age [18]. Both scoring manuals [19, 20] define deep sleep partly through an absolute amplitude criterion, which means an amplitude criterion learned on one derivation and one age band need not transfer. Second, classifier outputs trained under one label prior are miscalibrated under another, and the correction is well characterised [21, 22]. These observations motivate the class-wise transfer analysis in Section 4.5; because cohort and derivation change together, they are treated as candidate explanations rather than mechanisms identified by this design.

3. Materials and methods

3.1. Data and labels

The Sleep Cassette subset of Sleep-EDF Expanded comprises 153 recordings from 78 subjects; we use the Fpz–Cz derivation at 100 Hz with 30 s epochs (3,000 samples per epoch). Annotations are mapped to W, N1, N2, N3 and REM; Movement and Unknown epochs retain their temporal position but are excluded from the loss and from all metrics. After windowing, 195,767 epochs remain, of which 195,469 are scored and 298 are Movement/Unknown.

An evaluation window is applied identically to both cohorts. It is anchored on the first and last true sleep epoch (any of N1, N2, N3, REM) and extended by 30 min of wake on each side, i.e. 60 epochs per side. This policy is stated explicitly because it strongly affects the wake fraction and hence every aggregate metric, and because it is one of the principal reasons published numbers are not directly comparable. It is a benchmark window, not a deployable inference rule, since its anchors use ground-truth labels.

The SHHS Visit-1 sample was locked before any prediction was inspected and comprises 180 test participants, with a further 5 adaptation and 15 validation participants that were never used to update weights. The EEG channel is C4–A1 at 125 Hz, resampled to 100 Hz with a polyphase anti-aliasing filter (up 4, down 5, Kaiser window $\beta = 5.0$) applied to the continuous signal before epoching. SHHS labels are mapped $0 \rightarrow W$, $1 \rightarrow N1$, $2 \rightarrow N2$, $\{3, 4\} \rightarrow N3$, $5 \rightarrow REM$, $\{6, 9\} \rightarrow \text{excluded}$. Table 1 reports both label distributions.

The prior shift between cohorts is substantial: total variation distance 0.187, $D_{KL}(EDF \parallel SHHS) = 0.101$ nats. SHHS1 contains proportionally $2.02\times$ as much N3 and $0.38\times$ as much N1. We report the per-class ratios in Table 1 because Section 4.5 uses their *direction* as evidence.

3.2. Preprocessing conditions

All conditions operate on the continuous calibrated signal in microvolts before epoching. Four active signal conditions enter the performance analysis, and one audit-only condition (E5) records a planned clipping ablation:

- **Raw** (E0–E2): no filtering, no clipping, no rescaling; native μV retained.
- **Band-pass only** (E4): zero-phase Butterworth band-pass, order 4, 0.5–30 Hz, implemented as second-

Table 1. Scored-label distribution in the two evaluation samples.

Stage	Sleep-EDF		SHHS1 test		
	Epochs	Share	Epochs	Share	Ratio to EDF
W	65,941	33.73%	36,983	21.88%	0.65
N1	21,522	11.01%	7,002	4.14%	0.38
N2	69,132	35.37%	76,287	45.14%	1.28
N3	13,039	6.67%	22,806	13.49%	2.02
REM	25,835	13.22%	25,934	15.34%	1.16
Total scored	195,469	100%	169,012	100%	

Table 2. Experimental configuration codes. The identifiers denote signal–model combinations, not a performance ranking or a sequence of model versions.

ID	Signal condition	Epoch encoder	Sequence model and role
E0	Raw	15-branch CNN	BiLSTM; internal control
E1	Raw	15-branch CNN	TCN; isolates the sequence model
E2	Raw	ResNet-1D	TCN; replaces E1 feature/context package with current-epoch encoder
E3	Band-pass + clip + fixed scale	ResNet-1D	TCN; fixed-scale configuration
E4	Band-pass only	ResNet-1D	TCN; effect of filtering alone
E5	Band-pass + clip	ResNet-1D	TCN; audit only, identical input to E4
E6	Band-pass + clip + z-score	ResNet-1D	TCN; alternative rescaling

order sections with forward–backward filtering.

- **Band-pass plus clipping** (E5; audit only): the E4 signal followed by a hard clip at $\pm 800 \mu\text{V}$. Across all 153 Sleep-EDF recordings, E5 and E4 were bitwise identical and the measured clipped-sample fraction was zero. Fold 0 was retained as an audit artifact; the remaining folds and all performance tests were cancelled before scores were inspected because E5 did not define an independent input condition.
- **Fixed-scale amplitude handling** (E3): the same band-pass, followed by a hard clip at $\pm 800 \mu\text{V}$ and division by a constant factor of 100. The clip threshold is an absolute constant identical for every recording, so no record-level or cohort-level statistic is estimated and cross-subject leakage is not introduced; relative amplitude between recordings is preserved.
- **Per-record z-score** (E6): the same band-pass and clip, followed by subtraction of the mean and division by the standard deviation computed over the whole filtered, clipped recording. This uses no target labels but does use all samples of the target recording, removes absolute amplitude differences between recordings and is transductive at the recording level rather than purely inductive.

No notch filter is applied: 50 Hz lies outside the 0.5–30 Hz pass-band and at the Nyquist frequency of the 100 Hz signal. The contrast E3 – E6 therefore evaluates a practical preprocessing choice: preserving calibrated cross-recording amplitude relationships or applying record-wise normalisation. Because clipping never activated in E5, E3 and E4 differ in these data by the fixed division by 100; this identity is used only to interpret the observed contrast and does not establish equivalence or a hardware-independent scaling rule.

3.3. Model configurations

Six active configurations (E0–E4 and E6) enter performance analysis; E5 documents the cancelled clipping-only contrast. The control encoder produces 75 features from 15 CNN branches spanning current, previous and next epochs (C/P/N). ResNet-1D produces 128 features from the current epoch only. The non-causal six-block TCN has a 253-epoch receptive field; the BiLSTM control has 128 hidden units per direction. Both sequence models process the full recording with masked padding. Layer dimensions and all hyperparameters

are reported in Supplementary Methods S1.

3.4. Two-stage training and subject-wise cross-validation

The 78 subjects are partitioned into 10 subject-disjoint folds. In each evaluation round one fold is the test fold, the next fold (modulo 10) is the validation fold and the remainder are training folds. Both nights of a subject always share the same role. The same partition is used for every configuration, and the test fold never participates in model selection; the training API accepts only training and validation loaders, so test data cannot enter fitting.

Training has two stages. The epoch encoder is fitted with class weights calculated from training subjects, selected on validation data, hash-verified and used to cache features without gradients. The sequence model is then fitted on the frozen cache using masked cross-entropy and selected by validation macro-F1. E1 reuses E0 encoder checkpoints, so E1 – E0 isolates the sequence-model substitution. In contrast, E2 – E1 changes the encoder, removes explicit P/N inputs and changes the feature dimension from 75 to 128; it is a bundled feature/context replacement. Optimisers, learning rates, batch sizes and early-stopping settings are in Supplementary Methods S1.

3.5. Test locking, statistical analysis and effect sizes

All six configurations completed validation-based selection before the test set was opened, once. In Sleep-EDF, pooled macro-F1 is the descriptive headline metric, but the unit of statistical inference is the subject. Paired subject-cluster bootstrap with 10,000 resamples estimates confidence intervals for the difference in pooled macro-F1, while a two-sided Wilcoxon signed-rank test [23] assesses the subject-level distribution of per-subject differences [24]. Holm correction [25] is applied over the four pre-specified comparisons E1 – E0, E2 – E1, E3 – E2 and E3 – E6. These two procedures target different estimands—a pooled, epoch-weighted quantity and a subject-averaged rank quantity—and may therefore disagree; where they do, we say so explicitly rather than reporting only the favourable one. E3 – E0 is labelled post-hoc throughout.

We also report win/tie/loss counts, sign dominance $(W - L)/(W + L)$ and an unadjusted exact sign test as descriptive robustness checks; they are not part of the pre-specified inferential family.

After the primary results had been observed, the whole campaign was repeated with a second fixed training seed on the same partition. The two seeds are analysed separately; p -values are never pooled, and two fixed seeds are not treated as a sample from the distribution over initialisations. This repeat is reported as a post-protocol sensitivity analysis.

For SHHS, each configuration uses all ten Sleep-EDF outer-fold checkpoints. Softmax probabilities are averaged in float64 across folds before the final argmax for each epoch. The pre-locked primary criterion is subject-mean macro-F1, with E3 – E0 and E3 – E6 as the two locked comparisons under the same bootstrap and Wilcoxon–Holm treatment. The E1/E2 component analysis is secondary evidence on an already-opened sample, and E3 – E2 is explicitly post-hoc.

3.6. Operational evaluation

Inference timing was measured on identical hardware (Tesla V100) for all configurations and excludes I/O, preprocessing and training. Each measurement used a batch of one synthetic 100-epoch sequence, 20 warm-up passes and three rounds of 100 timed forward passes. Configuration order was shuffled in each round; we report the median over all 300 passes. This benchmark measures the implemented full forward pipeline for one fixed tensor shape, not overnight wall-clock inference or algorithmic complexity.

4. Results

4.1. Out-of-fold performance on Sleep-EDF

E3 is descriptively highest on all four metrics (Table 3). A pooled ranking is not by itself statistical evidence, and Table 4 shows how little of this ordering survives inference.

Only E3 – E6 is consistent across all three lenses: a fully positive bootstrap interval, Holm significance, and a sign test that agrees. The remaining pre-specified contrasts have intervals containing zero and do not

Table 3. Out-of-fold test performance on Sleep-EDF (seed 42, pooled over all folds).

Configuration	macro-F1	Accuracy	κ	F1 (N1)
E0	0.7754	0.8262	0.7614	0.5009
E1	0.7802	0.8315	0.7683	0.5115
E2	0.7835	0.8341	0.7716	0.5180
E3	0.7904	0.8363	0.7754	0.5276
E4	0.7891	0.8360	0.7746	0.5270
E6	0.7691	0.8230	0.7573	0.5124

Table 4. Pre-specified paired comparisons on Sleep-EDF macro-F1 ($n = 78$ subjects). Holm correction is over the four pre-specified rows; sign-test statistics are unadjusted supplementary robustness checks. E3 – E0 is post-hoc and is excluded from the Holm family.

Comparison	Δ macro-F1	CI _{95%}	p_{Holm}	W/T/L	Δ_{sign}	p_{sign}
E1 – E0	0.004811	[−0.000915, 0.010193]	0.1022	51/0/27	+0.308	0.0088
E2 – E1	0.003251	[−0.002370, 0.008808]	0.1036	44/0/34	+0.128	0.3082
E3 – E2	0.006962	[0.000305, 0.014520]	0.8989	37/0/41	−0.051	0.7343
E3 – E6	0.021319	[0.012179, 0.030698]	0.0012	49/0/29	+0.256	0.0308
<i>E3 – E0 (post-hoc)</i>	0.015024	[0.005746, 0.025871]	—	49/0/29	+0.256	0.0308

reach significance.

E3 – E2 deserves particular attention because it is the clearest case of estimand disagreement in the study. Its pooled bootstrap interval marginally excludes zero ([0.000305, 0.014520]), yet only 37 of 78 subjects improve, $\Delta_{\text{sign}} = -0.051$, and both the Wilcoxon and sign tests are far from significance ($p_{\text{sign}} = 0.734$). Pooled macro-F1 is epoch-weighted and is therefore dominated by subjects contributing more scored epochs; the subject-level statistics weight every subject equally. When these disagree, the honest reading is that the effect is carried by a subset of high-epoch-count subjects and is not a general property of the population. We do not claim E3 improves on E2 in-domain.

For the post-hoc E3 – E0 contrast, the added macro-F1 of the complete pipeline is 0.0150 with a fully positive interval; because it changes architecture and preprocessing simultaneously, it supports the pipeline as a whole and cannot attribute credit to any component.

4.2. Per-class and seed-sensitivity checks

Detailed per-class metrics, the Sleep-EDF confusion matrix and both seed-sensitivity tables are provided in Supplementary Tables S1–S4. In domain, N1 is E3’s hardest class and the largest E3–E6 F1 difference is on N3 (+0.0628). All four pre-specified differences remain positive under seed 123, but only E3 – E6 has a fully positive bootstrap interval in both runs; its Holm-adjusted result is significant for seed 42 only and its magnitude falls from 0.0213 to 0.0102. Thus, the direction is stable across these two fixed seeds, but the inferential strength is not; two seeds are not an independent replication or an estimate of initialisation variance.

4.3. Complexity, latency and training cost

As Table 5 shows, the ResNet-1D–TCN family provides a measured 3.76 $\times$ speed-up over the CNN–BiLSTM control while carrying 4.37 $\times$ the parameters and 28.4% more peak allocated memory. The implementation is therefore faster but not smaller. E1 differs from E0 only in the sequence model and improves latency by just 1.07 $\times$; the larger E2–E0 difference is consequently associated primarily with replacing 15 independently executed CNN branches by one ResNet-1D encoder, together with the full-pipeline implementation. The benchmark does not isolate kernel-launch overhead or predict latency for other sequence lengths and hardware.

Table 5. Full-pipeline forward benchmark on a Tesla V100 (fold 0, seed 42; batch 1, sequence length 100). Throughput is in epochs s^{-1} and peak allocated memory in MiB; I/O and preprocessing are excluded.

ID	Parameters	Median (ms)	Throughput	Peak mem.	Speed-up
E0	248,630	13.4315	7,445	58.81	1.00×
E1	640,950	12.5111	7,993	60.31	1.07×
E2	1,085,578	3.5750	27,972	75.54	3.76×
E3	1,085,578	3.5687	28,021	75.54	3.76×
E6	1,085,578	3.5756	27,967	75.54	3.76×

In the sequential seed-123 campaign, complete training and validation for E2 and E3 was observed to be 12.2× and 10.2× faster than E0. E1 reused E0 encoder checkpoints and is not an end-to-end runtime comparison. Full timings are in Supplementary Table S5; they are workflow measurements under one scheduling and early-stopping regime, not hardware-independent complexity claims.

4.4. Cross-cohort evaluation on SHHS1

The seed-123 campaign preserves the direction and Holm significance of both locked SHHS contrasts, whereas the in-domain E3 – E6 comparison no longer reaches the Holm threshold. This sensitivity result uses the same SHHS sample and is not an independent cohort replication (Supplementary Table S6).

Both locked comparisons (Table 7) are unambiguous, and the sign-based effect sizes are large ($\Delta_{\text{sign}} = +0.533$ and $+0.389$; exact sign tests $p = 3.8 \times 10^{-13}$ and 1.9×10^{-7}). Supporting metrics move together: relative to E0, E3 gains 0.0338 pooled macro-F1, 0.0368 accuracy and 0.0461 κ . Within ± 1 epoch of a transition, E3 gains 0.0366 over E0 and 0.0182 over E6.

The component analysis is more equivocal. E1 – E0 remains positive but small, supporting the TCN substitution. E2 – E1 is *negative* out of domain (-0.0128 , interval entirely below zero), so the bundled replacement of E1’s 75-dimensional C/P/N package by the current-epoch 128-dimensional ResNet-1D representation does not generalise better on unprocessed SHHS signals. This contrast cannot attribute the reversal to the encoder alone. The post-hoc E3 – E2 contrast is the largest observed configuration difference ($+0.0475$, $\Delta_{\text{sign}} = +0.633$, 147/180 subjects improved). Because E3 and E2 share an architecture and differ only in preprocessing, this observed post-hoc comparison identifies preprocessing as the stronger development axis in this sample; it is not confirmatory evidence for a particular operation.

A seed-123 extension evaluating E4 (band-pass only) provides a more specific secondary contrast. E4 reached subject-mean macro-F1 0.5732, pooled macro-F1 0.6147, accuracy 0.7064, $\kappa = 0.5869$ and N1 F1 0.3492, exceeding same-seed E2 by 0.0323 and E3 by 0.0098. The E5 audit established that the $\pm 800 \mu V$ clip never activated, so E4 and E3 differ in this dataset by the fixed division by 100. The extension therefore associates band-pass filtering with most of the E4–E2 gain and provides no evidence that the additional fixed scaling improves transfer. Because the SHHS sample was already open and training stochasticity is represented by only two fixed seeds, these comparisons are secondary; they are not equivalence, non-inferiority or universal scaling tests.

Table 6. Cross-cohort performance on 180 locked SHHS1 participants (seed 42). No weights were updated on SHHS. E0 and E3 are inductive; E6 uses complete-record, unlabelled target statistics and is transductive. Intervals are subject-cluster bootstrap intervals for subject-mean macro-F1.

ID	Subject-mean macro-F1 (CI _{95%})	Pooled macro-F1	Accuracy	κ	F1 (N1)	Transition macro-F1
E0	0.5268 [0.5101, 0.5431]	0.5761	0.6648	0.5339	0.2998	0.4323
E3	0.5680 [0.5503, 0.5850]	0.6099	0.7016	0.5801	0.3451	0.4689
E6	0.5407 [0.5227, 0.5584]	0.5732	0.6742	0.5408	0.3102	0.4507

Table 7. Locked paired comparisons on SHHS1 subject-mean macro-F1 ($n = 180$).

Comparison	Role	Δ	CI _{95%}	p_{Holm}	W/T/L	Δ_{sign}
E3 – E0	locked primary	0.0412	[0.0314, 0.0512]	2.65×10^{-13}	138/0/42	+0.533
E3 – E6	locked primary	0.0274	[0.0182, 0.0370]	1.87×10^{-8}	125/0/55	+0.389

4.5. Which errors should target-domain adaptation prioritise?

E3’s pooled macro-F1 falls from 0.7904 in domain to 0.6099 out of domain, a gap of 0.1805. An aggregate gap does not indicate whether adaptation requires broad retraining or a targeted correction. We therefore localise the loss by class and confusion channel. This is a prioritisation analysis: it identifies where an intervention could recover the most performance, but does not by itself identify the physiological or acquisition mechanism that generated the errors. Throughout this analysis, $A \rightarrow B$ denotes a reference-labelled A epoch predicted as B; it does not denote a physiological stage transition over time.

N3 is the dominant class loss: E3 N3 F1 falls from 0.8085 on Sleep-EDF to 0.4055 on SHHS1, with precision 0.9440 but recall 0.2582. E0 shows nearly the same recall (0.2610) and $N3 \rightarrow N2$ rate (72.3% versus 73.1% for E3), so E3’s aggregate transfer advantage does not resolve N3 and the failure cannot be assigned specifically to ResNet-1D or TCN. $N2 \rightarrow \text{REM}$ is the second-largest E3 confusion channel.

The E3 oracle diagnostic ranks $N3 \rightarrow N2$ and $N2 \rightarrow \text{REM}$ as potential adaptation targets, corresponding to 74.5% and 27.5% of the numerical cross-cohort gap. These nonlinear values are neither additive nor achievable-performance estimates. Reference-label boundary neighbourhoods are harder on both cohorts, but low N3 recall in stable SHHS1 regions prevents a boundary-only explanation. These are post-hoc offline diagnostics, not physiological-transition or causal analyses.

E6 provides a label-free but transductive sensitivity check. Its complete-target-record z -score lowers N3 recall to 0.2005 and leaves boundary-neighbourhood recall essentially unchanged (0.0721 versus 0.0733 for E3). Thus, the tested normalisation is not a sufficient stand-alone remedy. Detailed class-wise, confusion, oracle and transition results are in Supplementary Tables S7–S11.

5. Discussion

5.1. Principal findings

The first research question asked whether the sequence-model substitution and bundled E2 feature/context replacement justified changing the pipeline. They did not establish a stable predictive advantage: both in-domain configuration effects were small and non-significant after multiplicity correction, and the bundled E2–E1 contrast reversed sign on SHHS1. The supported pipeline benefit is operational: the implemented ResNet-1D–TCN pipeline reduced measured forward latency and observed training-and-validation wall-clock time, while increasing parameter count and peak GPU memory. It did not establish superior predictive accuracy.

The second question asked which choices retained benefit across cohorts. The fixed-scale E3 procedure outperformed its two locked references on SHHS1, and E3 ranked above E6 under both seeds. Under a fixed ResNet-1D–TCN architecture, the post-hoc E3–E2 contrast was larger than the E1–E0 sequence-model contrast and the bundled E2–E1 feature/context contrast; the seed-123 E4 extension and the E5 identity audit

associate band-pass filtering with most of this difference and provide no evidence that fixed scaling adds a transfer benefit. These secondary contrasts identify preprocessing as the more promising development axis, but they do not establish a causal or universal contribution for any single operation.

The third question asked where adaptation should focus. $N3 \rightarrow N2$ recurs at essentially the same rate in E0 and E3, while the E3 oracle analysis ranks it above $N2 \rightarrow \text{REM}$ in potential leverage; the N3 deficit becomes extreme near reference-label changes but remains substantial in stable N3. The evidence therefore identifies a target-domain failure shared across the evaluated pipelines, not an E3-specific weakness or a boundary-only phenomenon. Record-wise z -scoring does not repair N3, so a labelled, class-specific target-domain adaptation experiment is better motivated than another generic architecture or normalisation change.

5.2. From error localisation to an adaptation decision

The high precision and low recall of N3, the concentration of N3 errors in N2, and the further recall loss near stage transitions are consistent with a conservative N3 decision boundary in SHHS1. E0’s near-identical N3 recall and $N3 \rightarrow N2$ rate show that this failure persists across the two model families evaluated; they do not identify a shared causal mechanism. Montage and age are plausible contributors: the model was trained on Fpz–Cz in a younger cohort and evaluated on C4–A1 in an older cohort, while slow-wave amplitude varies with both derivation and age [17, 18]. However, montage, population, acquisition and scoring practice change together between the datasets. The present design cannot assign the failure to one of them, and the precision–recall pattern alone does not identify an amplitude threshold as the mechanism.

The difference in class prevalence is also relevant. A prior-shift correction would move N3 and N1 in the required directions, but it was not fitted or validated here. It should therefore be evaluated as a low-cost baseline in the next target-domain experiment, alongside class-specific calibration and limited fine-tuning; the present error pattern does not justify dismissing it or treating it as a complete solution.

E6 provides an important boundary on the interpretation. If removal of record-level absolute scale were a sufficient correction, E6 should improve N3 recall over E3. Instead, E6 lowers whole-sample N3 recall from 0.2582 to 0.2005 and leaves transition-region recall essentially unchanged (0.0721 versus 0.0733). Thus, the data do not support absolute amplitude scale as a sufficient explanation, and record-wise z -scoring is not a useful stand-alone adaptation strategy for this failure.

The actionable result is narrower but better supported. Before deployment on SHHS-like data, the next experiment should target the $N2$ – $N3$ decision boundary using labelled target-domain examples and should report N3 recall specifically near transitions. A small calibration or fine-tuning study on the held-out SHHS adaptation participants would test whether the boundary can be moved without increasing false N3 predictions. The $N2$ – REM channel should be the second target. This prioritisation follows directly from the error counts; the choice between montage-aware features, class-specific calibration and target-domain fine-tuning remains an open experimental question.

5.3. Architecture and preprocessing implications

E1–E0 isolates the TCN substitution; E2–E1 jointly changes the encoder, explicit P/N context and feature dimension and cannot establish a ResNet-only effect. Neither in-domain contrast survives Holm correction, and the bundled E2–E1 contrast reverses on SHHS1. The reliable advantage of the implemented ResNet-1D–TCN pipeline is therefore operational: a $3.76\times$ benchmark speed-up and 10.2 – $12.2\times$ observed training-and-validation speed-up, paid for with more parameters and memory. In the observed post-hoc E3–E2 comparison, preprocessing is the stronger development axis. The E4 extension associates band-pass filtering with much of that difference, but the opened cohort and two fixed seeds preclude a causal or universal claim about any operation.

5.4. Positioning relative to published results

We do not claim state-of-the-art benchmark performance. The wake-trimming policy alone (Section 3.1) moves the wake fraction, and hence accuracy and macro-F1, by an amount comparable to every architectural effect measured here; subject-count, split scheme, and channel choice compound this. Published Sleep-EDF

figures for the models discussed in Section 2 [5, 6, 7, 2] were obtained under protocols that differ in several of these dimensions simultaneously, so a numerical comparison would misrepresent the evidence in either direction. Our E0 control is a re-implementation of the ZleepAnlystNet design evaluated under *our* protocol, and it is the reference for the component questions posed here. The study should therefore be read as evidence about relative design choices and transfer failure under a shared protocol, not as a leaderboard claim.

5.5. Limitations and a prioritised agenda

Results should be read within the data and protocol evaluated. SHHS1 is one specific out-of-domain sample, not a population. The second seed corroborates direction but two fixed seeds cannot characterise initialisation variance; a proper treatment would require enough seeds to estimate a distribution, and we report no such claim. Several component and reference-transition analyses were run after the sample was opened and are supporting evidence only. Single-channel EEG limits separability for N1 and REM in principle, and the transition neighbourhood uses reference labels on both sides of a boundary, so it is an offline benchmarking device rather than a deployable rule. No probability-calibration or selective- prediction analysis was performed, which matters for clinical use where an abstain option would be valuable [22, 7].

Two reporting gaps remain. Absolute-performance intervals are available for the SHHS configurations (Table 6) but not for the Sleep-EDF ones, where only difference intervals were retained. And although paired win/tie/loss counts and median subject-level differences were preserved for every comparison, the full per-subject macro-F1 distributions were not archived, so we cannot report the dispersion (IQR, range) of individual performance or identify which participants the models handle worst. For a metric whose clinical relevance is inherently per-patient this is a limitation. Per-class SHHS metrics have now been recovered for E6 from the locked prediction artifacts. Pooled class-wise results for E0 have also been recovered from the locked run manifest and support the cross-model N3 comparison; transition-neighbourhood and stable-interior metrics are available for E0 and E3 on both cohorts. Corresponding class-wise results for E1 remain unavailable.

Ordered by expected practical value, we would prioritise: (1) limited class-specific fine-tuning or decision-boundary adjustment on the held-out SHHS adaptation participants, evaluated on N3 recall and false-positive rate in both boundary and stable regions; (2) fitting and evaluating calibration/selective prediction on independent target validation data; and (3) estimating initialisation variance with a sufficient number of seeds. The first two address the deployment failure identified here; the third improves the strength of the methodological evidence.

6. Conclusion

A shared subject-wise protocol allowed six active configurations to be compared on the same Sleep-EDF folds; a seventh planned configuration, E5, was excluded before performance analysis because it was bitwise identical to E4. The isolated TCN substitution and bundled E2 feature/context replacement did not establish a stable predictive advantage after correction for multiple comparisons. ResNet-1D–TCN nevertheless offers an operational alternative that is $3.76\times$ faster in the controlled forward benchmark and $10.2\text{--}12.2\times$ faster in observed training-and-validation wall-clock time, at the cost of more parameters and memory.

On SHHS1, the fixed-scale E3 procedure outperformed both locked references, while the largest observed component difference was associated with preprocessing under a fixed architecture. This result shifts the practical priority from further architecture replacement to signal handling and target-domain evaluation, without establishing any individual preprocessing operation as causal.

Most importantly, the aggregate transfer loss resolves into concrete adaptation targets. E0 and E3 both systematically under-call N3 as N2, showing that E3’s overall transfer advantage does not solve N3 detection; N2 is also frequently over-called as REM by E3. Per-record z -scoring does not repair the N3 deficit, and the cross-cohort diagnostic shows that reference-label boundaries intensify errors without fully explaining the stable-region SHHS1 deficit. Thus, neither the evaluated architecture changes nor generic normalisation is a sufficient deployment remedy. The evidence supports a targeted, labelled adaptation study on these decision boundaries; it does not yet support an architecture-specific, physiological or clinical claim.

Reproducibility and data availability

Source code, experimental protocols, data partitions, run manifests and aggregate analysis outputs are retained in the project repository. Sleep-EDF Expanded is publicly available from PhysioNet. SHHS is accessed through the NSRR under its data use agreement; participant-level derived data are not re-distributed. Checkpoints are verified by SHA-256 before loading, and the locked SHHS predictions used for the reported analyses were hash-verified against their run manifests. The E6 class-wise results are therefore reported as a descriptive reanalysis of these locked predictions. The historical locked-protocol snapshot `configs/shhs_zero_shot_v1.json` is retained in the repository and its SHA-256 matches the protocol hash recorded in the SHHS run manifest ('165d7cdf...fe93'). The newer `configs/shhs_v1_protocol.json` has a different SHA-256 ('9541e233...fe9') because it is an expanded post-run audit record; it must not be substituted for the historical snapshot when interpreting the locked predictions. Protocol provenance is therefore traceable, although the locked E6 results remain a descriptive reanalysis rather than a rerun.

Ethics

This study is a secondary analysis of de-identified data from research repositories with their own governance mechanisms; no new participants were recruited. Use of SHHS follows NSRR access conditions. All interpretations apply to the samples and protocols described and do not constitute clinical validation.

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CRediT authorship contribution statement

Ngo Nhat Quan: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Visualization, Writing – original draft, Writing – review and editing. Pham Thai Son: Supervision, Methodology, Validation, Writing – review and editing. Both authors approved the final manuscript.

Declaration of competing interests

The authors declare no competing interests.

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